



Beryl

△ Fine, octagonal step-cut aquamarine with excellent clarity

Beryl provides some of nature’s most beautiful gemstones. Although it is colourless in its pure form, it is perhaps best known for its coloured varieties, which include aquamarine and emerald – indeed, its name comes from the Greek *beryllos*, meaning “green stone”. The colourless form of beryl is known as goshenite, and its clarity is such that it was used to make lenses for some of the earliest eyeglasses during the late Middle Ages.

The colours of beryl

Where colours do occur in beryl, they are caused by minute chemical impurities, and this is sometimes reflected in the varieties’ names. The green colour of emerald, for example, is caused by traces of chromium. Morganite is coloured pink, rose lilac, peach, orange, or pinkish yellow by the presence of manganese, and its crystals sometimes show colour banding, with a sequence from blue near the base to nearly colourless in the centre, to peach or pink at the tip. It is almost always faceted, and stones with a yellow or orange tinge are sometimes heat-treated to emphasize their pink tones. Manganese is also the colouring agent in the rare red beryl, sometimes called red emerald or scarlet emerald. The colours in blue and green aquamarine (meaning “sea water”), and yellow to golden heliodor (from the Greek *helios*, meaning “sun”), result from traces of iron. Much greenish-blue aquamarine is heated to produce an intense blue colour that has become popular in modern jewellery.

Key pieces



Fabergé egg | This fabulous egg was made by Carl Fabergé (see pp.278–79). It is crafted from gold, platinum, and silver, and is set with aquamarines and diamonds and holds a gold model of a cruiser.



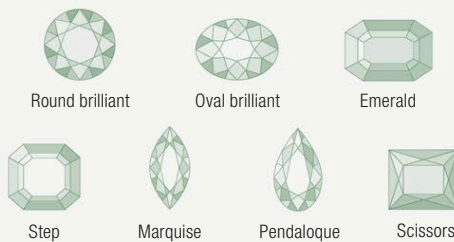
Cartier clip brooch | Designed by Cartier in 1935, the platinum and diamond setting of this clip brooch holds an unusual late 17th-century or early 18th-century Indian carved emerald in the form of a flower.



Aquamarine brooch | Made in 1967 by British jeweller and goldsmith John Donald, this gold brooch features an unusually cut aquamarine – part faceted, part cabochon. The stone is ringed with a gold setting consisting of sections of tubing.

Specification

Chemical name Beryllium aluminium silicate | **Formula** $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ | **Colours** Colourless, red, blue, green, yellow
Structure Hexagonal | **Hardness** 7.5–8 | **SG** 2.6–2.8
RI 1.57–1.60 | **Lustre** Vitreous | **Streak** White



Locations
1 USA **2** Colombia **3** Brazil **4** Ireland **5** Norway **6** Sweden
7 Germany **8** Austria **9** South Africa **10** Zambia
11 Mozambique **12** Madagascar **13** Russia